Lipids: fatty acids and triacylglycerols

Lipids constitute a chemically heterogeneous group of compounds with a common feature, their **poor solubility in water**.

Biological functions of lipids

The biological functions of lipids are as diverse as their chemical diversity:

- **Storage function:** lipids can store a large amount of energy in a very compact way thanks to their hydrophobic nature.
- **Structural function:** lipids are the main structural component of biomembranes.
- **Cofactors for enzymes:** for example, vitamin K (which regulates blood clot formation) or coenzyme Q (involved in ATP synthesis in mitochondria).
- **Metabolic regulators and signaling molecules**: some hormones and growth factors have a lipidic nature.
- **Antioxidant function**: tocopherols, such as vitamin E, are biological antioxidants that react with reactive oxygen species and other free radicals, protecting biomolecules from oxidation.
- **Others:** bile acids help in the absorption of fat and fat-soluble vitamins, subcutaneous fat helps to keep body warmth, etc.

Classification of lipids

Based on their chemical nature, lipids are classified as:

- **Simple lipids**: esters of fatty acids with glycerol or other alcohols.
- **Compound lipids**: esters of fatty acids containing other groups in addition to alcohol. They can be phospholipids or glycolipids.
- **Derived lipids**: a heterogeneous group containing isoprenoids, fat-soluble vitamins, steroids, prostaglandins, etc.

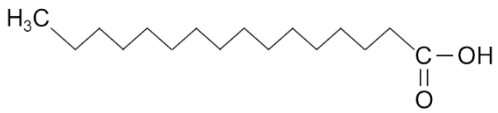
Based on their function, lipids are classified as:

- **Storage lipids:** the primary storage lipids found in animals are triacylglycerols. They are composed of a glycerol molecule and three fatty acid molecules, which can be released and utilized as energy when needed.
- **Membrane lipids:** they form the bilayers of the cell membranes. The three major classes of membrane lipids are phospholipids, glycolipids and cholesterol.

- **Biologically active lipids:** heterogeneous group of lipids present in the body in smaller amounts than storage or structural lipids. They include steroid hormones (derived from **cholesterol**), signaling molecules (such as arachidonic acid) and lipid-soluble vitamins (many of them derived from **isoprene**).

Fatty acids

Fatty acids are the most common component of lipids. Chemically, they are **carboxylic acids with an aliphatic chain** containing 4 to 36 carbon atoms.



The most common natural fatty acids are **unbranched** and contain an **even** number of carbons between 12 and 24. The even number of carbons is a consequence of their mechanism of synthesis, which takes place through the **condensation of two-carbon acetate units**. Solubility of fatty acids decreases as the number of carbon atoms increases.

The carbon atoms in fatty acids are numbered as C1, C2, C3, etc., starting from the COOH group. Alternatively, they can be numbered starting from the methyl group as ω1, ω2, ω3, etc.

Fatty acids can be **saturated** or **unsaturated**. Saturated fatty acids lack carbon double bonds. Unsaturated fatty acids can be **monounsaturated** (one double bond) or **polyunsaturated** (with two or more double bonds).

Saturated fatty acids

They have the formula $CH_3–(CH_2)_n–COOH$. They are named by adding the suffix *-anoic* after the hydrocarbon name. They can also be named by the number of carbon atoms and the number of double bonds (in this case 0) separated by a colon. Finally, they also have a common name.

For example:

- Hexadecanoic acid » 16:0 » palmitic acid
- Octadecanoic acid » 18:0 » stearic acid

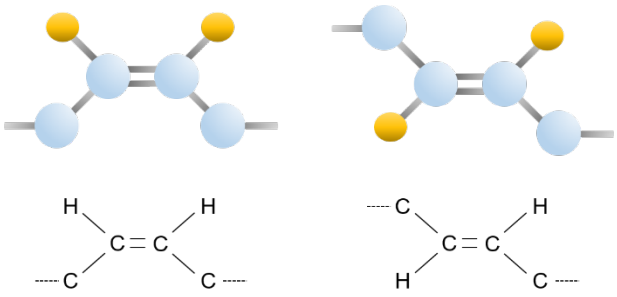
Unsaturated fatty acids

They are named by adding the suffix *-enoic* after the hydrocarbon name preceded by the type and position of the double bond(s). As for saturated fatty acids, they are also named by the carbon skeleton or by a given name.

For example:

- *cis*-9-Octadecenoic acid » $18:1\Delta^9$ » oleic acid

Double bonds in natural unsaturated fatty acids are commonly found in ***cis* configuration**.



Cis configuration *Trans* configuration

In most monosaturated fatty acids the double bond is between C9 and C10. In polyunsaturated fatty acids (PUFA), double bonds are typically separated by a methylene group (−CH_2−).

Unsaturated fatty acids have lower melting points compared to saturated fatty acids of the corresponding size due to their reduced ability to pack together.

Two fatty acids are essential for humans (i.e. we need them, but our metabolism cannot synthesize them):

- Linoleic acid (18:2$\Delta^{9,12}$)
- α-Linolenic acid (18:3$\Delta^{9,12,15}$)

Both fatty acids have to be supplied in the diet.

Another important PUFA is arachidonic acid (20:4$\Delta^{5,8,11,14}$), precursor of prostaglandins, thromboxanes and leukotrienes.

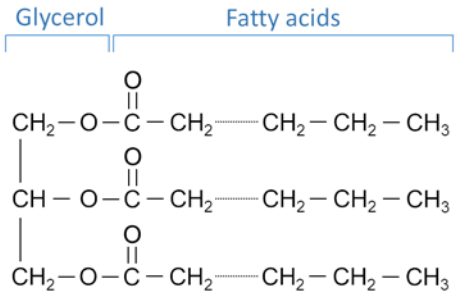
Clinical and epidemiological studies suggest that PUFA with a double bond between the third and fourth carbon starting from the methyl end (**ω3 PUFA**) have a potential role against cardiovascular disease. This association may be related to the role of these fatty acids in inflammation, platelet aggregation, hypertension or hyperlipidemia. The "Mediterranean diet" is rich in ω3 PUFA.

T*rans* fatty acids

Trans fatty acids are formed by partial hydrogenation of unsaturated fatty acids. They are present in dairy and bakery products, because partially dehydrogenated vegetable oils are used in their cooking (which increases the stability of the oils at high temperatures and give foods a better taste and texture). *Trans* double bonds allow the fatty acid to adopt a more extended configuration. In that way, *trans* fatty acids can pack more regularly and show higher melting points than *cis* forms. Intake of *trans* fats has been associated with an increased risk of diseases such as heart disease or diabetes.

Triacylglycerols

Triacylglycerols (TAG) contain **three fatty acids esterified to the three hydroxyl groups of glycerol**. They are also known as triglycerides and neutral fats.



The carbon atoms of glycerol are designated as α, β and γ, or as 1, 2 and 3. Naturally occurring TAGs usually contain two or three different fatty acids. Each animal species has a characteristic pattern of fatty acid composition. Human body fat contains 50% oleic acid, 25% palmitic acid, 10% linoleic and 5% stearic acid. A typical TAG in humans is the *1-stearoyl, 2-linoleoyl, 3-palmitoyl glycerol*.

Because the polar hydroxyl groups of glycerol and the polar carboxylate groups of fatty acids are forming the ester bonds, TAGs are **nonpolar hydrophobic water-insoluble molecules**.

Oils are TAGs that contain a high proportion of unsaturated or short-chain fatty acids. Oils are generally found in plants and are liquid at room temperature. **Fats** contain more saturated and long-chain fatty acids and, consequently, are solid at room temperature. Fats are mainly of animal origin.

In our body, TAGs are predominantly found in **adipocytes**. The main function of TAGs is to store energy and provide insulation.

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